

KAIRAV DUTTA

BTech CSE @ IIIT Delhi '29 · Builder · Systems Programmer · Full-Stack Developer

[✉ kairavdutta@gmail.com](mailto:kairavdutta@gmail.com) | [🐙 github.com/ka1rav6](https://github.com/ka1rav6) | [💼 linkedin.com/in/kairavdutta](https://www.linkedin.com/in/kairavdutta) | 📍 Delhi, India | [🌐 ka1rav6.github.io](https://ka1rav6.github.io)

PROFILE

18-year-old CS undergrad at IIIT Delhi who builds things that ship. Equally at home writing a C rule engine with a bytecode VM and shipping a React + Django full-stack app. Systems programming, graphics, AI/ML, language design — if it compiles, runs, and solves a real problem, I'm in. JEE Advanced AIR 5217. 24+ projects and counting.

EDUCATION

IIIT Delhi — BTech in Computer Science

Jul 2025 – Expected 2029

CBSE Class XII — 97.4% (Centum in Mathematics)

Delhi Private School, Dubai · 2024–2025

CBSE Class X — 97.6%

PD Model School

Age: 18 | **Languages:** 8+ (C, C++, Python, TS, Java, etc.) | **Projects:** 24+ | **JEE Adv. Rank:** AIR 5217 | **Orgs:** Byld · LDA · IEEE · Cultural Council

TECHNICAL SKILLS

Languages: C (C11/C23), C++ (C++17/20), Python, TypeScript, JavaScript, Java, HTML/CSS, x86-64 Assembly

Frontend & UI: React 18/19, Vite, Tailwind CSS, JavaFX, Streamlit, ImGui, EzUI

Backend & DB: FastAPI, Flask, Django REST, Node.js, PostgreSQL, MongoDB, SQLite

Systems & Graphics: OpenGL, GLFW, SDL2, Qt 5/6, Linux Sockets, ASIO, CMake/Make

AI / ML: PyTorch, MediaPipe, OpenCV, scikit-learn, NumPy/Pandas, OR-Tools

Tools & Publishing: VS Code Extensions, Git/GitHub, pybind11, vcpkg, Maven, Pygame, PyPI

EXPERIENCE

Dev Team Member | May 2026 – Present

Byld — IIIT Delhi

Designing and shipping tech products at a student-run organisation. Full-stack development and product design.

Operations Team Member | Sep 2025 – Present

LDA IIITD / Cultural Council / IEEE

Managing logistics and operations for the Literary & Dramatics Association at IIIT Delhi.

PROJECTS

Vela — Rule Engine

Parses JSON rules into bytecode, evaluates via stack-machine VM. Arena allocator, bitmask facts, semantic checker.

C · yyjson · bytecode VM

EzUI — Reactive C++ GUI [\[VIBECODED\]](#)

Minimal reactive C++ GUI wrapping SDL2 with declarative API, reactive state, HStack/VStack layout.

C++20 · SDL2

K Language — Custom Transpiler

Custom programming language transpiling to C++. Lexer → Parser → Transpiler. Cleaner syntax, low-level control.

Python · C++

LazyCMake — CMake TUI [\[VIBECODED\]](#)

lazygit-inspired TUI for creating, configuring & building CMake C/C++ projects. No hand-writing CMakeLists.txt.

C++20 · FTXUI · CMake

HTTP Server (Raw Sockets)

HTTP/1.1 server from scratch using Linux sockets. Raw TCP/IP, request parsing, static file serving.

C++ · Linux Sockets

College LMS MVP

Google Classroom clone — role-based (Teacher/TA/Student), 25+ API endpoints, file upload, responsive UI.

Django REST · React · PostgreSQL

CTF Game (3D) [In Progress]

3D Capture-The-Flag with OpenGL, Assimp model loading, ReactPhysics3D physics, ImGui debug UI.

C++20 · OpenGL · Physics

Variable Case Converter [VIBECODED]

VS Code extension — converts camelCase, PascalCase, snake_case, kebab-case across 12+ languages. Marketplace.

TypeScript · VS Code API

Header File Generator [VIBECODED]

VS Code extension — auto-generates C/C++ header files from source. Extracts prototypes, classes, structs.

TypeScript · VS Code API

Textmex — Terminal Editor

Vim-like terminal text editor with gap buffer, raw terminal I/O, cursor movement, file saving.

C++ · Terminal I/O

pykworldsim — ECS Simulation [VIBECODED]

Production-grade ECS world sim: people, cities, relationships, jobs, goals. Deterministic RNG, snapshot/replay. PyPI.

Python · ECS · PyPI

Orbix Migraine — AI Predictor

Predicts migraines from sleep, hydration, stress, screen time. Probabilistic model with adaptive learning.

Python · FastAPI · scikit-learn

Tetris (C++/SDL2)

Fully playable Tetris with piece movement/rotation, collision detection, line clearing, OOP design.

C++ · SDL2 · CMake

Gesture Recognition Game

Hand gesture recognition for a game using MediaPipe, OpenCV, PyTorch with CUDA acceleration.

Python · MediaPipe · PyTorch

Efficient Data Engine

C++ core + Python CLI + Tkinter GUI. Matrix ops, stats, sorting, CSV. pybind11 bridge.

C++ · Python · pybind11

ROTA Gen — Shift Scheduler

Employee shift scheduler using OR-Tools CP-SAT constraint programming. Color-coded Excel output.

Python · OR-Tools · openpyxl

Burnout Risk Predictor

ML web app predicting student burnout/dropout risk from academic & psychological data. Random forest.

Python · scikit-learn · Streamlit

Chess Engine (C)

Board/piece abstractions, move parsing, game state management, debug logger.

C11 · CMake

BEYOND CODE

 Chess School chess society |  Reading Avid reader |  Game Dev C++/SDL2, OpenGL, Pygame |  Systems Low-level, VMs, compilers |  AI/ML PyTorch, MediaPipe |  Open Source Published VS Code extensions & PyPI packages

Full portfolio: ka1rav6.github.io